

MTH 439: Abstract Algebra Notes

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Chapter 1

Rings

1.1 Rings and Subrings

We now continue to generalize the familiar arithmetic operations of addition and multiplication. Just as with groups, results from the algebra of arithmetic will be extended to a host of non-numerical objects, e.g., matrices, functions, polynomials, etc.

Definition 1.1. A *ring* is a nonempty set, say R , with two binary operations (usually written as multiplication and addition)¹ that satisfies the following:

1. If $a, b \in R$ then $a + b \in R$. (closed under addition)
2. $a + (b + c) = (a + b) + c$ (associative addition)
3. $a + b = b + a$ (commutative addition)
4. There exists an 0_R such that $a + 0_R = 0_R + a = a$ (additive identity)
5. There exists an $-a \in R$ such that $a + (-a) = 0_R$ for every $a \in R$. (additive inverse)
6. If $a, b \in R$ then $ab \in R$. (closed under multiplication)
7. $a(bc) = (ab)c$ (associative multiplication)
8. $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ (distribution holds).

- Note that parts 1-5 say that a ring is an *abelian group*.²

Example 1.1. The set of integers, $\mathbb{Z}$, under addition and multiplication is a ring.

Example 1.2. $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ ($\mathbb{Z}$ modulo 4) is also a ring under addition and multiplication as in $\mathbb{Z}_n$ for any $n \in \mathbb{N}$.

Example 1.3. The set of 2 by 2 matrices with real numbers as entries,

$$\mathcal{M}_2(\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}.$$

¹But of course the operations are not necessarily addition and multiplication. $+$ and $\times$; $\oplus$ and $\otimes$; are among the different notations that may be used in varying texts.

²Commutative groups are also called abelian in honor of Norwegian mathematician Niels Abel.

Closure, associativity, and distribution can be shown using the usual rules of matrix multiplication and addition. $\mathcal{M}_2(\mathbb{R})$ is commutative under addition (but not multiplication), the additive inverse is $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$, the additive identity is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, and the multiplicative identity is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Proposition 1.1. *If a and b are two elements in R then $(-a)(-b) = ab$.*

Proof. By definition $a + (-a) = 0$. $(a + (-a))(-b) = 0$ implies $a(-b) + (-a)(-b) = 0$ by distribution. The additive inverses of $a(-b)$ is unique thus $a(-b) + (-a)(-b) = 0$ implies that $(-a)(-b) = ab$. $\square$

- In particular for rings like $\mathbb{Z}$ and $\mathbb{R}$, the product of two negatives is a positive. Note the importance of distribution.

Exercise 1.1. Show that $\mathbb{Z}_6$ is a ring with unity under addition and multiplication.

Exercise 1.2. Show that the set $\sqrt{2}\mathbb{Z} = \{x: x = m + \sqrt{2}n \text{ for some } m, n \in \mathbb{Z}\}$ is a ring.

Exercise 1.3. Let R and S be rings. Define addition and multiplication on the Cartesian product $R \times S$ by,

$$(r, s) + (r', s') = (r + r', s + s') \text{ and } (r, s)(r', s') = (rr', ss')$$

for $r, r' \in R$ and $s, s' \in S$. Prove that $R \times S$ is a ring.

Definition 1.2. A *commutative ring* is a ring R such that $ab = ba$ for all $a, b \in R$.

Definition 1.3. A ring with *identity*³ is a ring R with an element 1_R such that $a1_R = 1_Ra$ for all $a \in R$.

- It is important to check that a potential 1_R is both a *right identity*, $a1_R = a$; and a *left identity*, $1_Rb = b$.

Example 1.4. $\mathbb{C}$, $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{Z}$ are all commutative rings with identity under the usual addition and multiplication.

Example 1.5. $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ where $n \geq 2$ is a commutative ring with identity under the usual addition and multiplication (recall that n is called the *modulus*).

- To add $x + y \pmod n$ find $(x + y) \div n$ and find the remainder, say r , then $(x + y) \pmod n = r$. In the same way we can find $(xy) \pmod n$. For example, $(3 \cdot 7) \pmod 8 = 5$ since $2 \cdot 8 + 5 = 3 \cdot 7$ (divide by 8 and take the remainder).

Proof. Since $r < n$, $\mathbb{Z}_n$ will be closed under addition and multiplication. Parts 2, 3, 4, and 7 hold because of the normal rules of arithmetic. Which also give multiplicative commutativity. 0 and 1 are the additive and multiplicative identities respectively.

Which only leaves distribution, part 8, and additive inverse, part 5. Let $a \in \mathbb{Z}_n$ then $a < n$ so then $n - a \in \mathbb{Z}_n$, and $a + (n - a) = 0$. For distribution, let $a, b, c \in \mathbb{Z}_n$. $a(b + c) \pmod n = (ab + ac) \pmod n = ab \pmod n + ac \pmod n$ (this was shown last semester). $\square$

Example 1.6. $E = \{x: x = 2n \text{ for some } n \in \mathbb{Z}\}$, the set of even integers is a commutative ring. However it does *not* have identity.

³The term *unity* is often used in place identity. A ring with an identity element is said to “have unity”.

- Keep in mind that a ring is *not* a group under multiplication. Elements in a ring do not necessarily have a multiplicative inverse, and rings do not necessarily have a multiplicative identity. Keep examples of rings in mind which do not have these properties.

Exercise 1.4. Prove that a ring that is cyclic⁴ under addition and commutative (under multiplication).

Theorem 1.1. *If a ring has an identity, it is unique.*

Proof. Suppose to the contrary that a ring R has two identity elements 1 and $1'$. $a1 = a$ for all $a \in R$ including $1' \in R$. Thus $1' \cdot 1 = 1'$ and $1 \cdot 1' = 1$. By the definition of a ring $1' = 1 \cdot 1' = 1' \cdot 1 = 1$. Therefore $1' = 1$, and the multiplicative identity is unique. $\square$

Exercise 1.5. Let F be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Define addition and multiplication as

$$(f + g)(x) = f(x) + g(x) \text{ and } (fg)(x) = f(x)g(x).$$

Show that F is a commutative ring with identity.

Exercise 1.6. If a ring element has a multiplicative inverse, it is unique.

1.2 Subrings

Definition 1.4. Let R be a ring and S be a subset of R .⁵ If S is a ring under the same operations as R , then S is a *subring*.

Example 1.7. $\mathbb{Q}$ is a subring of $\mathbb{R}$, and $\mathbb{Z}$ is a subring of $\mathbb{Q}$ and $\mathbb{R}$.

Example 1.8. The set of even integers is a subring of $\mathbb{Z}$, but the set of odd integers is not since it is not closed under addition.

The following two theorems provide shortcuts when trying to show that a subset is a subring. Particularly, The second theorem can save quite a bit of work.

Theorem 1.2. *Suppose that R is a ring and $S \subseteq R$ such that the following hold:*

1. *If $a, b \in S$ then $a + b \in S$ (closed under addition)*
2. *If $a, b \in S$ then $ab \in S$ (closed under multiplication)*
3. $0_R \in S$.
4. *If $a \in S$ then there is an $-a \in S$ such that $a + (-a) = 0_R$.*

Then S is a subring of R .

Proof. Parts 2, 3, 7, and 8 of the Ring definition hold for all elements in R since it is a ring. So they hold for all elements in S . The remaining requirements 1-4 in the theorem satisfy the remaining requirements. $\square$

- Rings and thus subrings must be nonempty. A requirement which is covered by 3. Since $0_R \in S \Rightarrow S \neq \emptyset$.

⁴Recall that a group G is *cyclic* if for every $x \in G$ there is some $g \in G$ such that $x = g + g + \cdots + g$. In which case we write $G = \langle g \rangle$ and say that G is *generated* by g , e.g., $\mathbb{Z}_3 = \langle 1 \rangle = \langle 2 \rangle$ since $2 + 2 = 1$, $2 = 2$, and $2 + 2 + 2 = 0 \pmod{3}$.

⁵Notated: $S \subseteq R$.

Theorem 1.3. *A nonempty subset S of a ring R is a subring if whenever $a, b \in S$:*

1. $ab \in S$ (closed under multiplication)
2. $a - b \in S$ (closed under subtraction)

Proof. If $a - b \in S$ for all $a, b \in S$ then S is a subgroup of R (subgroup test), and since R is additively commutative it follows that S is an abelian subgroup of R under addition (this also means that $0_R \in S$). S is multiplicatively and additively associative; and S is distributive over addition because these properties hold in R . $\square$

Exercise 1.7. Show that the set $3\mathbb{Z} = \{x: x = 3n \text{ for some } n \in \mathbb{Z}\}$ is a subring of $\mathbb{Z}$.

Exercise 1.8. Show that the set $k\mathbb{Z} = \{x: x = kn \text{ for some } n \in \mathbb{Z}\}$ for any integer k is a subring of $\mathbb{Z}$.

Exercise 1.9. Prove that the intersection of any two subrings of a ring R is also a subring of R .

1.3 Integral Domains and Fields

For $\mathbb{Z}_n$ it is easy to see that $ab = 0$ does *not* imply that a or b is 0. For example, in $\mathbb{Z}_6$ $2 \cdot 3 \bmod 6 = 6 \bmod 6 = 0$. Of course many rings (indeed the most familiar rings) do have this property. In $\mathbb{R}$, if $x \cdot y = 0$ then $x = 0$ or $y = 0$.

Definition 1.5. An *integral domain* is a commutative ring R with identity such that $1_R \neq 0_R$ ⁶ and if $ab = 0_R$ then $a = 0_R$ or $b = 0_R$ for any $a, b \in R$.⁷

Example 1.9. $\mathbb{Z}$ is an integral domain.

Let $a, b \in \mathbb{Z}$. Suppose that $b \neq 0$ (if $b = 0$ then the condition is met) then $ab = 0 \Rightarrow a = 0/b = 0$.

Example 1.10. The ring $\mathbb{Z}[x]$, the polynomials with integer coefficients, is an integral domain.

Exercise 1.10. Show that $\mathbb{Z}_6$ is *not* an integral domain.

Exercise 1.11. Prove that $\mathbb{Z}_n$ is not an integral domain if n is not prime.

Hint If n is not prime then $n = mq$ for two integers $1 < m, q < n$. Looking at the example seen in exercise might help.

Exercise 1.12. Prove that $\mathbb{Z}_p$ is an integral domain if p is prime.

Exercise 1.13. Prove that $\mathbb{Q}$ is an integral domain.

Exercise 1.14. Prove that the ring from exercise 1.5, F , is not an integral domain.

Hint Here are two *indicator functions*

$$\mathbf{1}_{[0, \infty)}(x) = \begin{cases} 1 & \text{if } x \in [0, \infty) \\ 0 & \text{if } x \notin [0, \infty) \end{cases} \text{ and } \mathbf{1}_{(-\infty, 0]}(x) = \begin{cases} 1 & \text{if } x \in (-\infty, 0] \\ 0 & \text{if } x \notin (-\infty, 0] \end{cases}$$

The indicator function is a function that has an output of either 1 or 0. For example, $\mathbf{1}_{[0, \infty)}(2) = 1$ and $\mathbf{1}_{[0, \infty)}(-2) = 0$. Now consider the functions $f(x) = x\mathbf{1}_{[0, \infty)}$ and $g(x) = x\mathbf{1}_{(-\infty, 0]}$.

⁶The condition $1_R \neq 0_R$ is necessary so that the ring $\{0\}$ is not an integral domain.

⁷If a, b are nonzero elements in a ring and $ab = 0$ then a and b are called *zero divisors*. Thus an integral domain is a commutative ring with identity that has no zero divisors.

Example 1.11. The ring of 2×2 matrices with real entries, $M_{2 \times 2}(\mathbb{R})$, is not an integral domain.

Proof. This ring has zero $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ so the matrices $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ are nonzero and $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. $\square$

When two nonzero elements in a ring have a product that is 0, we call the elements *zero divisors*. Having no such elements allows for cancellation by the following theorem.

Theorem 1.4. Let R be an integral domain and let $a, b, c \in R$ where $c \neq 0$. If $ac = bc$, then $a = b$.

Proof. Since rings have distribution, $ac = bc \Rightarrow ac - bc = 0 \Rightarrow (a - b)c = 0$. Because R is an integral domain $a - b = 0$ which implies that $a = b$. $\square$

Note that we could not prove theorem 1.4 using division, i.e., $\frac{ac}{c} = \frac{bc}{c}$, because we cannot assume that $c^{-1} \in R$. For example, $3x = 3y \Rightarrow x = y$ in $\mathbb{Z}$ but $\frac{1}{3} \notin \mathbb{Z}$. While $\mathbb{Z}$ is an integral domain, it has no element x such that $3x = 1$. That is, 3 (and all other numbers except 1) does not have a multiplicative inverse. All the elements in $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ (except 0) do have multiplicative inverses. Thus $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are all examples of the following definition.

Exercise 1.15. Let a and b be two nonzero elements in a ring R . Prove that in general, $ax = b$ does not have a unique solution.

Hint It should be understood that x must be in R .

Definition 1.6. A *field* is a commutative ring R with identity $1_R \neq 0_R$ ⁸ if for every $a \neq 0_R$ in R , there exists an $a^{-1} \in R$ such that $aa^{-1} = 1_R$.

- Sets $\supseteq$ Groups $\supseteq$ Rings $\supseteq$ Commutative Rings with identity $\supseteq$ Integral Domains $\supseteq$ Fields

Example 1.12. $\mathbb{R}$ and $\mathbb{Q}$ are fields.

Example 1.13. The set of *constructable numbers*, numbers which can be constructed with a straight edge and compass, is a field.

Exercise 1.16. Prove that $\mathbb{C}$ is a field.

Theorem 1.5. $\mathbb{Z}_p$ is a field if p is a prime.

Proof. This is left as an exercise. $\square$

Theorem 1.6. Cancellation. Let a, b, c be elements in an integral domain. Suppose that $a \neq 0$. Then $ab = ac \Rightarrow b = c$.

Proof. Let $a \neq 0$ for elements a, b, c in an integral domain, R . $ab = ac \Rightarrow a(b - c) = 0_R$. Because R is an integral domain either $a = 0$ or $b - c = 0$. By assumption, $a \neq 0$ so then $b - c = 0 \Rightarrow b = c$. $\square$

Theorem 1.7. A field is an integral domain.⁹

Proof. This is left as an exercise. $\square$

⁸Once again, this assumption is needed to that $\{0\}$ is excluded from this class.

⁹Often a *field* is defined as an integral domain in which every nonzero element has an multiplicative inverse.

Theorem 1.8. *A finite integral domain is a field.*

Proof. Let R be a finite integral domain with identity. Let $a \in R$ and $a \neq 0_R$. All we need to show is that a has an inverse. If $a = 1_R$ then $a^{-1} = 1_R$. Now suppose that $a \neq 1_R$ (if no such exists then we are done). Since R is finite there are two integers such that $i < j$ and $a^i = a^j$ (otherwise R would be infinite). By cancellation (above theorem) we have that $a^{j-i} = 1_R$. Because $a \neq 1_R$ we know that $j - i > 1$. Thus $a^{j-i-1}(a) = 1_R$ as needed. $\square$

Exercise 1.17. Prove that a field is an integral domain.

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Definition Example Exercise Theorem
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Chapter 2

Homomorphisms and Isomorphisms

Consider the two groups $\mathbb{Z}_6$ and $\mathbb{Z}_2 \times \mathbb{Z}_3$ (under addition). $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ and

$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)\}.$$

These groups have the same number of elements, but as it turns out they have much more in common. Pair the elements as follows:

$$\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ (0,0) & (1,1) & (0,2) & (1,0) & (0,1) & (1,2) \end{array}$$

Table 2.1: The elements from $\mathbb{Z}_6$ are mapped to $\mathbb{Z}_2 \times \mathbb{Z}_3$.

This map identifies the zero and identities of each group, $0 \rightarrow (0, 0)$ and $1 \rightarrow (1, 1)$ respectively. Note that in $\mathbb{Z}_6$, $3 + 4 = 1 \pmod 6$ and in $\mathbb{Z}_2 \times \mathbb{Z}_3$, $(1, 0) + (0, 1) = (1, 1)$. Our map identifies $1 \rightarrow (1, 1)$. Our map can be formulated with $(3x + 4y) \pmod 6 \rightarrow (x, y)$ is a function from $f: \mathbb{Z}_6 \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$.

Though these groups appear quite different, this function shows that their behavior is identical. We will call this function an *isomorphism* and two such groups *isomorphic*.

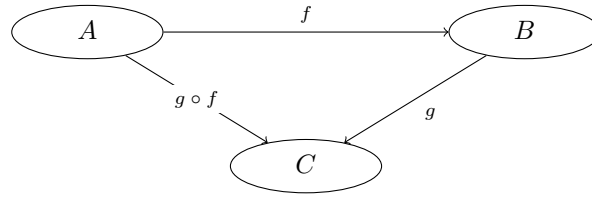
2.1 Functions

We covered functions in MTH 335, but that was a while ago.

Definition 2.1. A *function* (or *mapping*) from sets A to B , $f: A \rightarrow B$, is a rule that assigns each element in A to exactly one element in B .

- A is called the *domain* of f .
 - B is called the *range* of f .¹
 - If $f(a) = b$ for some $a \in A$, then b is called the *image of a under f*.
 - $\{b: b \in B \text{ and } f(a) = b \text{ for some } a \in A\}$ is called the *image of A under f*.
- Functions are well defined. Meaning that an input is assigned to one and only output.

¹Also called the *co-domain*.

Figure 2.1: Composition of functions f and g .

Example 2.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$. Then $f(x) = x^2 + 1$ is a function. it is clear from the equation that f is well defined.

Example 2.2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$. Then $f(x) = \pm\sqrt{x}$ is not a function. Since $f(1) = +1$ and -1 .

Example 2.3. Consider $f: \mathbb{R} \rightarrow \mathbb{Z}$ given by the *floor function* $f(x) = \lfloor x \rfloor$ where $\lfloor x \rfloor$ is the largest integer less than or equal to x , e.g., $\lfloor \pi \rfloor = 3$. Note that the floor function is also a function from $\mathbb{R}$ to $\mathbb{R}$, $f: \mathbb{R} \rightarrow \mathbb{Z}$.

Example 2.4. Let $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x, y) \rightarrow xy$. Then f is a function.

Example 2.5. Let $A = \{a, b, c, \dots, z\}$ and $f: A \rightarrow \mathbb{Z}$ be function defined by assigning the n^{th} letter to n , e.g., $f(3) = c$.

Definition 2.2. Let $f: A \rightarrow B$ and $g: B \rightarrow C$. The *composition* of $g \circ f$ is the mapping from A to C defined as follows: $g \circ f(a) = g(f(a))$ for any $a \in A$.

Example 2.6. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = 2x + 1$ and $g(x) = x^2$. Then $f \circ g(x) = f(g(x)) = 2(x^2) + 1 = 2x^2 + 1$.

Definition 2.3. A function, $f: A \rightarrow B$, is called *one-to-one*² if $f(a_1) = f(a_2)$ implies that $a_1 = a_2$.

Example 2.7. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ where $f(x) = x^3$ is one-to-one.

Proof. Suppose that $f(x_1) = f(x_2)$ for some $x_1, x_2 \in \mathbb{Z}$. So $(x_1)^3 = (x_2)^3 \Rightarrow ((x_1)^3)^{1/3} = ((x_2)^3)^{1/3} \Rightarrow x_1 = x_2$ □

Exercise 2.1. Show that any linear function $f: \mathbb{R} \rightarrow \mathbb{R}$ with a non-zero defined slope is one-to-one.

Example 2.8. $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = |x|$ is not one-to-one.

$|-1| = 1 = |1|$ but $-1 \neq 1$ so f is not one-to-one.

Definition 2.4. A function $f: A \rightarrow B$ is *onto* if for every $b \in B$ there is an $a \in A$ such that $f(a) = b$.

Example 2.9. $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = x^2$ is onto.

Proof. If $b \in \mathbb{R}$ then $\sqrt{b} \in \mathbb{R}$, and $f(\sqrt{b}) = (\sqrt{b})^2 = b$. □

Example 2.10. $f: \mathbb{Z} \rightarrow \mathbb{R}$ where $f(x) = x^2$ is not onto.

Proof. $2 \in \mathbb{R}$ but $\sqrt{2} \notin \mathbb{Z}$. □

Exercise 2.2. Prove that the floor function (see example 2.3) is onto but not one-to-one.

²Also called an *injective function*

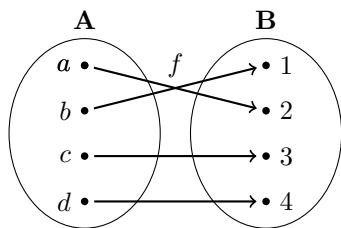


Figure 2.2:
 $f: A \rightarrow B$ is one-to-one and onto.

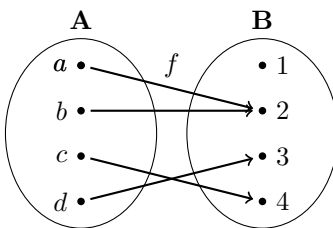


Figure 2.3:
 $f: A \rightarrow B$ is not one-to-one and is not onto.

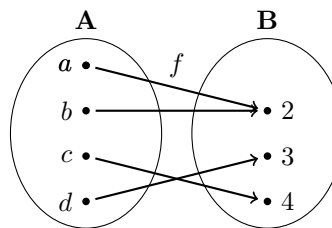


Figure 2.4:
 $f: A \rightarrow B$ is onto but not one-to-one.

2.2 Homomorphism

Definition 2.5. A (ring) homomorphism, f , from a ring R to a ring S is a function $f: R \rightarrow S$ that preserves the two ring operations:

$$f(a + b) = f(a) + f(b) \text{ and } f(ab) = f(a)f(b)$$

for $a, b \in R$.

Example 2.11. Consider the function $f: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ given by $f(x) = x^2$. f is a homomorphism.

$$f(a + b) = (a + b)^2 = a^2 + 2ab + b^2 = a^2 + b^2 \text{ since } 2ab = 0 \pmod{2}, \text{ and } f(ab) = (ab)^2 = a^2b^2 = f(a)f(b).$$

Exercise 2.3. Prove that $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$ given by $k \rightarrow k \pmod{n}$ is a ring homomorphism.

Example 2.12. The map $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x$ is not a homomorphism.

While the operation addition is preserved, $f(a + b) = 2(a + b) = 2a + 2b = f(a) + f(b)$, the multiplication operation is not, $f(ab) = 2ab \neq f(a)f(b) = 4ab$.

Exercise 2.4. Prove that the floor function (see example 2.3) mapping $\mathbb{R}$ to $\mathbb{Z}$ is not a homomorphism.

Exercise 2.5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2^x$. Show that f is not a homomorphism under the usual operations, but that is a homomorphism under addition and multiplication defined as follows: $2^x \oplus 2^y = 2^{x+y}$ and $2^x \otimes 2^y = 2^{xy}$.

Exercise 2.6. Prove that every ring homomorphism $f: \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ has the form $f(x) = ax$ where $a^2 = a$. Hint: look at the identity; call $f(1) = a$.

If $f: R \rightarrow S$ is a homomorphism and $g: S \rightarrow T$ is a homomorphism, is $g(f): R \rightarrow T$ a homomorphism? Prove your answer.

Exercise 2.7. Is the map $f: \mathbb{Z}_8 \rightarrow \mathbb{Z}_8$ given by $f(x) = 2x$ a ring homomorphism? Prove your answer.

Exercise 2.8. Is the mapping $f: \mathbb{Z}_5 \rightarrow \mathbb{Z}_{30}$ given by $f(x) = 6x$ a ring homomorphism? Prove your answer.

Theorem 2.1. Properties of Ring Homomorphisms. Let f be a ring homomorphism from rings R to S , and let A be a subring of R .

1. $f(0_R) = 0_S$.

2. $f(-a) = -f(a)$ for every $a \in R$.
3. $f(a - b) = f(a) - f(b)$ for every $a, b \in R$.
4. For any $a \in R$ and any positive integer n , $f(nr) = nf(r)$ and $f(r^n) = f(r)^n$.
5. If R is commutative then $f(R)$ is commutative.
6. $f(A) = \{f(a) : a \in A\}$ is a subring of S .
7. If R has unity 1_R , $S \neq \{0\}$, and f is onto
 - (a) then $f(1_R)$ is the unity of S .
 - (b) if $aa^{-1} = 1_R$ for any $a, a^{-1} \in R$, then $f(a)f(a^{-1}) = 1_S$ and $f(a)^{-1} = f(a^{-1})$.

Exercise 2.9. ★ Prove properties 1, 2, and 3 of theorem 2.1.

Exercise 2.10. ★ Prove properties 4 and 5 of theorem 2.1.

Exercise 2.11. ★ Prove property 6 of theorem 2.1.

Exercise 2.12. ★ Prove properties 7.a and 7.b of theorem 2.1.

2.3 Isomorphisms

Definition 2.6. A *ring isomorphism* from a ring R to a ring S is a function that is one-to-one and onto.

Definition 2.7. If there exists a ring isomorphism $f: R \rightarrow S$ then R and S are said to be *isomorphic*.

Showing that two rings are isomorphic usually requires finding an isomorphism between the, and this can be tricky at times. However looking at the properties from theorem 2.1 can serve as a guide.

Example 2.13. The function shown in table 2.1 is an isomorphism. The function $f: (x, y) \rightarrow (3x + 4y) \bmod 6$ is an isomorphism.

Example 2.14. The function $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$ from exercise 2.3 is not an isomorphism.

f is a homomorphism (exercise 2.3) and it is onto since $f(a) = a$ for any $a \in \mathbb{Z}_n$. However it is not one-to-one; consider $f(nk) = 0 \bmod n$ for any $k \in \mathbb{Z}$.

Exercise 2.13. Show that the rings $\mathbb{Q}$ and $\mathbb{R}$ are not isomorphic to $\mathbb{Z}$.

Exercise 2.14. Let $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$. Prove that $f: \mathbb{Q}[\sqrt{2}] \rightarrow \mathbb{Q}[\sqrt{2}]$ defined by $f(a + b\sqrt{2}) = a - b\sqrt{2}$ is an isomorphism.

Example 2.15. Prove that the $n\mathbb{Z}$ is ring isomorphic to $m\mathbb{Z}$ if and only if $m = n$. Recall that $n\mathbb{Z}$ is the ring of all n integer multiples.

Exercise 2.15. Show that any ring isomorphism, $f: \mathbb{Z} \rightarrow \mathbb{Z}$, must be the identity map.

Recall that a function $f: A \rightarrow B$ can be one-to-one only if $|A| \leq |B|$. That is, A having fewer elements than B is necessary condition for f to be one-to-one. Also, $|B| \leq |A|$ is a necessary condition for f to be onto. Clearly, $|A| = |B|$ is a necessary condition for f to be an isomorphism.